HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Investigated and resolved performance bottlenecks within complex Python-based AI infrastructure systems, reducing execution times by an average of **30%** through code profiling and optimisation.
- Developed and implemented performance improvements for compute-intensive workloads, focusing on enhancing memory usage and overall system throughput for machine learning model serving.
- Engineered Python scripts to automate data pipeline processes and scheduled reporting, ensuring timely delivery of insights for operational teams.
- Debugged and optimised existing codebases, identifying and fixing issues that impacted system performance and resource utilization across Linux environments.
- Collaborated with a team of 5 engineers on challenging technical problems, contributing to the design and implementation of performance-sensitive AI workloads.
- Built and maintained CI/CD pipelines using **Jenkins** and **Git** for automated testing and deployment of Python applications, improving deployment frequency by **40%**.
- Containerised Python applications using **Docker** and deployed them to **AWS EC2** instances, ensuring efficient resource management and scalability for production workloads.

SKILLS

Programming Languages – Python, C++, SQL, Java

Performance & Optimisation – Profiling, Benchmarking, Memory Management, Computational Efficiency, System Throughput

AI & ML Infrastructure – AI Workloads, Machine Learning Models, GPU-Accelerated Computing

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, Git, Linux

Data & Analytics – Data Pipelines, ETL, Pandas, NumPy, Scikit-learn

PROJECTS

Optimisation of Python ML Inference Service

- Refactored a Python-based service responsible for machine learning model inference, significantly improving response times and reducing CPU load by **22%**.
- Implemented profiling techniques to pinpoint performance bottlenecks in critical code paths, leading to targeted optimisations in data preprocessing and model marshalling.
- Deployed containerised service updates via **Jenkins** to **AWS**, ensuring stable rollouts and monitoring performance metrics post-deployment.

Performance Tuning for Compute-Intensive Data Pipelines

- Developed and tuned Python data pipelines for processing large datasets, enhancing efficiency and reducing processing time by approximately **35%**.
- Applied optimisation strategies to memory-intensive operations using **Pandas** and **NumPy**, enabling the handling of larger data volumes within existing resource constraints.
- Integrated automated performance testing into the pipeline's CI/CD workflow, providing continuous feedback on efficiency and stability.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)